Literature Review and Comparative Analysis

This document presents concise literature reviews of three IEEE papers related to UAV communication channels and adaptive modulation systems. A comparative analysis is also included to clearly explain how each paper is relevant to the proposed paper titled *“Adaptive Modulation Switching in 2D Wireless Communication Channels Under Varying System Parameters.”*

1. UAV-to-Ground Channel Modeling: (Quasi-)Closed-Form Channel Statistics and Manual Parameter Estimation

• Develops a 3D UAV-to-ground channel model for realistic wireless communication environments.

• Derives mathematical expressions for Doppler spectrum, fading behavior, Level Crossing Rate (LCR), and Average Fade Duration (AFD).

• Uses geometry-based stochastic modeling to represent UAV mobility and multipath propagation.

• Introduces a spatial-vector-based method to simplify channel statistics analysis.

• Studies the effect of mobility, elevation angle, and scattering on communication performance.

• Validates the proposed model using simulation and measurement results.

• Provides useful channel parameter estimation methods for practical UAV systems.

• Supports future 6G and MIMO-based UAV communication research.

2. A Survey of Air-to-Ground Propagation Channel Modeling for Unmanned Aerial Vehicles

• Provides a comprehensive review of UAV air-to-ground communication channel models.

• Discusses path loss, shadowing, Doppler effect, fading, multipath propagation, and antenna effects.

• Compares deterministic, stochastic, geometry-based, and ray-tracing channel models.

• Explains how UAV mobility and environmental conditions affect wireless communication.

• Analyzes the influence of carrier frequency, elevation angle, and scattering environments.

• Highlights challenges such as airframe shadowing, non-stationary channels, and interference.

• Reviews UAV applications in 5G, IoT, and future wireless networks.

• Emphasizes the importance of adaptive and reliable communication techniques.

3. Adaptive Modulation and Coding Technology in 5G System

• Studies Adaptive Modulation and Coding (AMC) techniques used in 5G communication systems.

• Uses channel state information (CSI) and SNR to dynamically select modulation schemes.

• Analyzes SNR-BLER performance for different modulation and coding schemes.

• Improves CQI mapping using region-fitting methods to reduce mapping error.

• Demonstrates how adaptive modulation improves throughput and spectral efficiency.

• Explains the use of QPSK, 16-QAM, 64-QAM, and higher-order modulation schemes.

• Combines AMC with OFDM and MIMO technologies for better wireless performance.

• Focuses on maintaining reliable communication under changing channel conditions.

Detailed Relevance and Similarity with Proposed Paper

Reference Paper	Relevance to Proposed Paper	Key Similarities with Proposed Paper
UAV-to-Ground Channel Modeling	This paper provides realistic UAV wireless channel models that can be used to extend the proposed adaptive modulation system into UAV communication environments. Its analysis of fading, Doppler effect, and mobility helps in understanding how channel variations influence modulation switching.	Both papers study wireless channel behavior under varying conditions. Both analyze fading, Doppler effects, mobility, and communication reliability. The proposed paper uses adaptive modulation to respond to changing channel conditions similarly addressed in this paper.
Survey of Air-to-Ground Propagation for UAVs	This survey provides theoretical background on UAV propagation environments, path loss, fading, and wireless impairments that are important for designing adaptive communication systems.	Both works focus on dynamic wireless communication environments and analyze how channel conditions affect communication quality. Both emphasize the importance of reliable and efficient communication techniques under varying SNR and fading conditions.
Adaptive Modulation and Coding in 5G	This paper is directly related to the proposed work because it studies SNR-based adaptive modulation and coding techniques used in modern wireless systems.	Both papers use SNR as the main parameter for modulation switching. Both aim to improve throughput, spectral efficiency, and communication reliability by dynamically selecting suitable modulation schemes under changing channel conditions.

The reviewed papers collectively establish the importance of adaptive wireless communication in dynamic environments. The first two papers provide strong foundations in UAV channel modeling and propagation analysis, while the third paper directly supports adaptive modulation concepts. The proposed paper combines these ideas by implementing an adaptive modulation switching framework in a 2D terrestrial wireless channel under varying system parameters such as SNR, fading, Doppler effect, path loss, and mobility.